NOTES:

1. MANUAL VALVES WITH POSITION FEEDBACK.
2. FLANGED ASME B16.5 CLASS 150.
3. TANK TO BE INSULATED.

SCHEME WATER DISTRIBUTION
2000-F24-PID-PR-10006
SCHEME WATER
13-0105-PSW-SS3-50

COOLING WATER DISTRIBUTION
2000-F13-PID-PR-10009
COOLING WATER RETURN
13-0116-COW-MSG1-250
25VB03S1 F13-10335
250VBFD1L1 F13-10336
25VB03S1 F13-10337
TO GRADE
TO GRADE

SCHEME WATER DISTRIBUTION
2000-F24-PID-PR-10006
SCHEME WATER
13-0043-PSW-SS3-50

VENT
DETAIL 9
HS 13418
XV 13418
50VB03S1 FL
LI 13426 HH H LL
LIT 13426
50VBD1S1 F13-10339
150VBFD1L2 F13-10379
F13-TK-409
COOLING WATER TANK 2
13-2224-COW-PEB-160
TO GRADE

VENT
O/F
F13-TK-408
COOLING WATER TANK 1
150VBFD1L1 F13-10340
13-0881-COW-MSG1-150
SP MS-006
TO GRADE

TIT 13420
TE 13420
TI 13420 H L
DN100
MSG1
SS1
250VBFD1L2 F13-10347
13-2203-COW-SS1-250-TI
SP MC-001
25VB03S1 F13-13930
250x200
F13-HE-420
CHILLER HEAT EXCHANGER NOTE 2
13-2205-COW-SS1-250-TI
25VB03S1 F13-13932
SP MC-001

TI 13421 H L
TIT 13421
TE 13421
DN100
SS1
MSG1
250VBFD1L2 F13-10350
13-0107-COW-MSG1-250-TI
COOLING WATER DISTRIBUTION
2000-F13-PID-PR-10009
COOLING WATER

13-1477-COW-SS1-250-TI
SP MC-001
25VB03S1 F13-13931
250VBFD1L2 F13-10349
TI 13427 H L
TIT 13427
TE 13427
DN100
DETAIL 9
HS 13419
XV 13419
50VB03S1 FL
25VB03S1 F13-10351
TO GRADE
VENT
O/F
TO GRADE
F13-TK-407
CHILLER TANK NOTE 3
LIT 13430
LI 13430 HH H L LL
50VBD1S1 F13-10351
300VBFD1L1 F13-10353
150VBFD1L1 F13-10352
13-0771-COW-PEB-160
TO GRADE

13-2204-COW-SS1-250-TI
TI 13437 H L
TIT 13437
TE 13437
DN100
SS1
MSG1
250VBFD1L2 F13-10348
25VB03S1 F13-13933
SP MC-001
13-0118-PSW-MSG1-300
13-0127-COW-MSG1-250-TI

ZIC 13433
ZSC 13433
ZSO 13433
ZIO 13433
NOTE 1
HV 13433
300VBFD1L1
13-0120-PSW-MSG1-300
SP ST-003
50VB03S1 F13-10354
50VB03S1 F13-10356
300x250
TO GRADE
F13-PW-416
CHILLER RECIRCULATING WATER PUMP 2
ZIC 13434
ZSC 13434
ZSO 13434
ZIO 13434
NOTE 1
HV 13434
250x200 250VCD1C1 F13-10358 250VBFD1L1
13-0121-PSW-MSG1-250

ZIC 13431
ZSC 13431
ZSO 13431
ZIO 13431
NOTE 1
HV 13431
300VBFD1L1
13-0118-PSW-MSG1-300
SP ST-003
50VB03S1 F13-10355
50VB03S1 F13-10357
300x250
TO GRADE
F13-PW-415
CHILLER RECIRCULATING WATER PUMP 1
ZIC 13432
ZSC 13432
ZSO 13432
ZIO 13432
NOTE 1
HV 13432
250x200 250VCD1C1 F13-10359 250VBFD1L1
FALL 13435
FSLL 13435
50VB03S1 F13-10360
TO GRADE

ZIO 13442
ZSO 13442
ZSC 13442
ZIC 13442
NOTE 1
HV 13442
13-0128-COW-MSG1-150-TI
150VBFD1L1
SP VC-003
150VCD1C1 F13-10365
200x150
25VB03S1 F13-10392
TO GRADE
F13-CU-402
CHILLER 1
VENDOR PACKAGE 2000-ME034-E02-10001
HS 13470
XI 13470A
XA 13470C
XI 13470B
START/ STOP
RUNNING
FAULT
READY
SP VC-003
200x150
25VB03S1 F13-10361
TO GRADE
DETAIL 9
HS 13441
XV 13441
150VBFD1L1 FL
TO GRADE

ZIO 13444
ZSO 13444
ZSC 13444
ZIC 13444
NOTE 1
HV 13444
13-0129-COW-MSG1-150-TI
150VBFD1L1
SP VC-003
150VCD1C1 F13-10366
200x150
25VB03S1 F13-10393
TO GRADE
F13-CU-403
CHILLER 2
VENDOR PACKAGE 2000-ME034-E02-10003
HS 13471
XI 13471A
XA 13471C
XI 13471B
START/ STOP
RUNNING
FAULT
READY
SP VC-003
200x150
25VB03S1 F13-10362
TO GRADE
DETAIL 9
HS 13443
XV 13443
150VBFD1L1 FL
TO GRADE

ZIO 13446
ZSO 13446
ZSC 13446
ZIC 13446
NOTE 1
HV 13446
13-0157-COW-MSG1-150-TI
150VBFD1L1
SP VC-003
150VCD1C1 F13-10367
200x150
25VB03S1 F13-10394
TO GRADE
F13-CU-404
CHILLER 3
VENDOR PACKAGE 2000-ME034-E02-10004
HS 13472
XI 13472A
XA 13472C
XI 13472B
START/ STOP
RUNNING
FAULT
READY
SP VC-003
200x150
25VB03S1 F13-10363
TO GRADE
DETAIL 9
HS 13445
XV 13445
150VBFD1L1 FL
TO GRADE

ZIO 13448
ZSO 13448
ZSC 13448
ZIC 13448
NOTE 1
HV 13448
13-0158-COW-MSG1-150-TI
150VBFD1L1
SP VC-003
150VCD1C1 F13-10368
200x150
25VB03S1 F13-10395
TO GRADE
F13-CU-405
CHILLER 4
VENDOR PACKAGE 2000-ME034-E02-10005
HS 13473
XI 13473A
XA 13473C
XI 13473B
START/ STOP
RUNNING
FAULT
READY
SP VC-003
200x150
25VB03S1 F13-10364
TO GRADE
DETAIL 9
HS 13447
XV 13447
150VBFD1L1 FL
250x150
13-0119-PSW-MSG1-250

13-0106-COW-MSG1-300-TI
TI
NOTE 1
ZSO 13422
ZSC 13422
ZIO 13422
ZIC 13422
HV 13422
SP ST-003
300VBFD1L1
50VB03S1 F13-10341
50VB03S1 F13-10343
300x250
TO GRADE
F13-PW-413
COOLING WATER PUMP 1
M
250x200 250VCD1C1 F13-10345 250VBFD1L1
NOTE 1
ZIO 13423
ZIC 13423
ZSO 13423
ZSC 13423
HV 13423
13-1381-COW-MSG1-250-TI
FSLL 13436
FALL 13436

13-0108-COW-MSG1-300-TI
TI
NOTE 1
ZSO 13424
ZSC 13424
ZIO 13424
ZIC 13424
HV 13424
SP ST-003
300VBFD1L1
50VB03S1 F13-10342
50VB03S1 F13-10344
300x250
TO GRADE
F13-PW-414
COOLING WATER PUMP 2
M
250x200 250VCD1C1 F13-10346 250VBFD1L1
NOTE 1
ZIO 13425
ZIC 13425
ZSO 13425
ZSC 13425
HV 13425
13-0109-COW-MSG1-250-TI

DRG No.	REFERENCES
2000-F00-PID-PR-10007	LEGEND SHEET 7 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10006	LEGEND SHEET 6 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10005	LEGEND SHEET 5 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10004	LEGEND SHEET 4 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10003	LEGEND SHEET 3 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10002	LEGEND SHEET 2 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10001	LEGEND SHEET 1 - PIPING AND INSTRUMENTATION DIAGRAM

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	05SEP25	GENERAL REVISIONS ADDED	AH	MM	LV	BS	DM
0	28OCT24	APPROVED FOR CONSTRUCTION	NB	JDM	LS	LS	DM

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: MILLING & CLASSIFICATION
MILL COOLING & CHILLER WATER CIRCUIT
PIPING AND INSTRUMENTATION DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NTS	2000-F13-PID-PR-10012	1

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